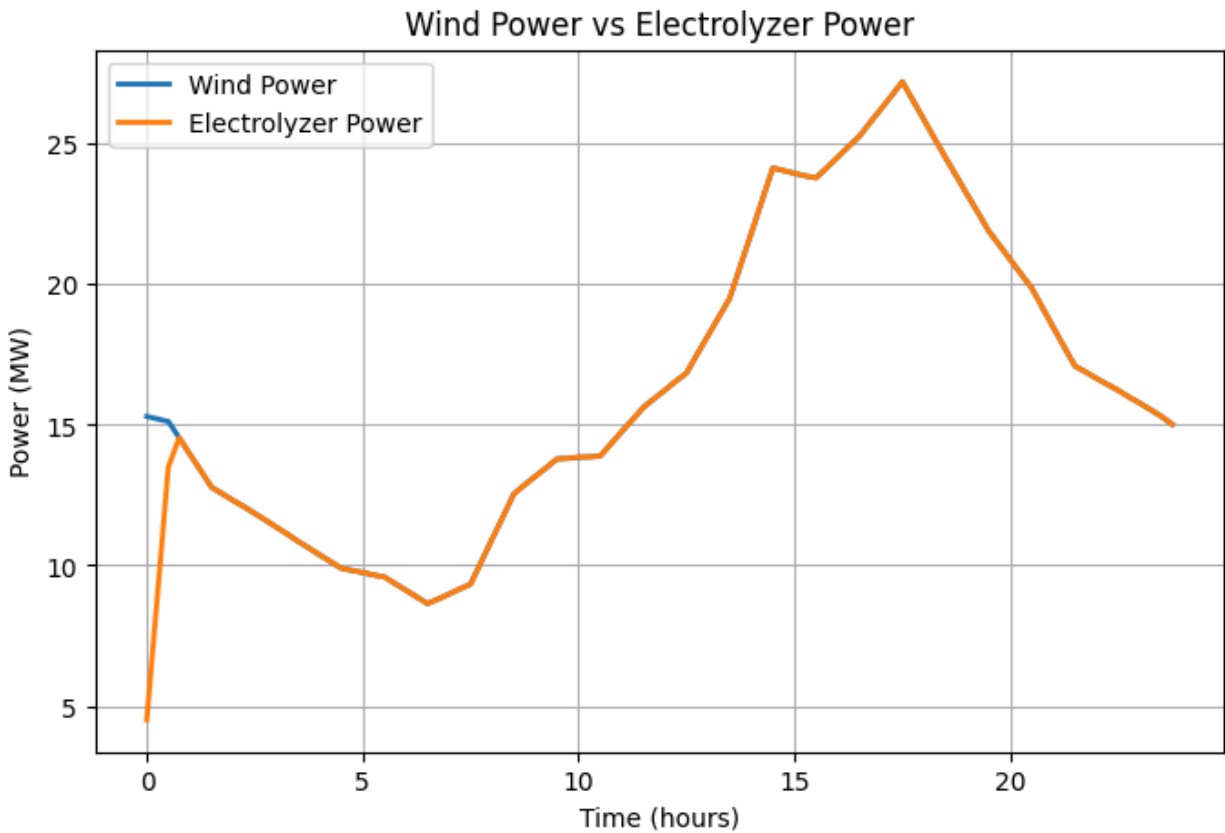
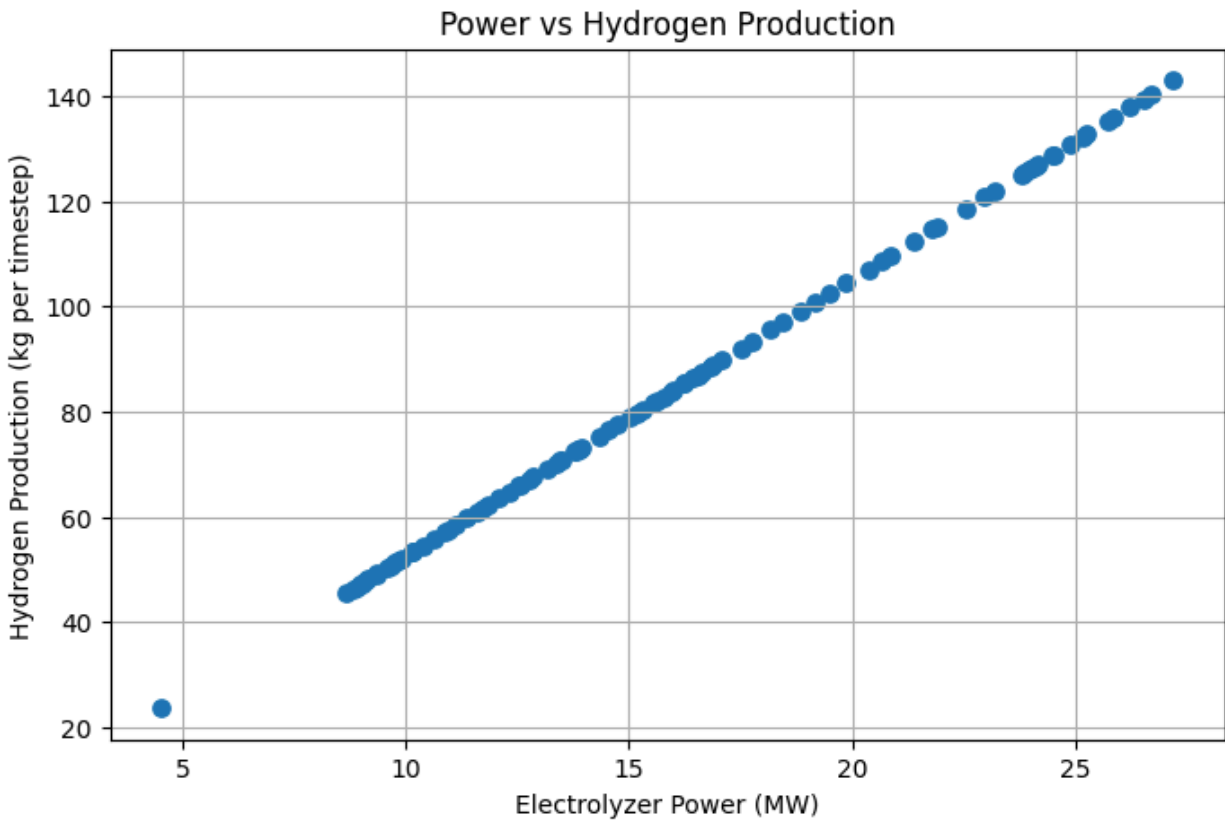


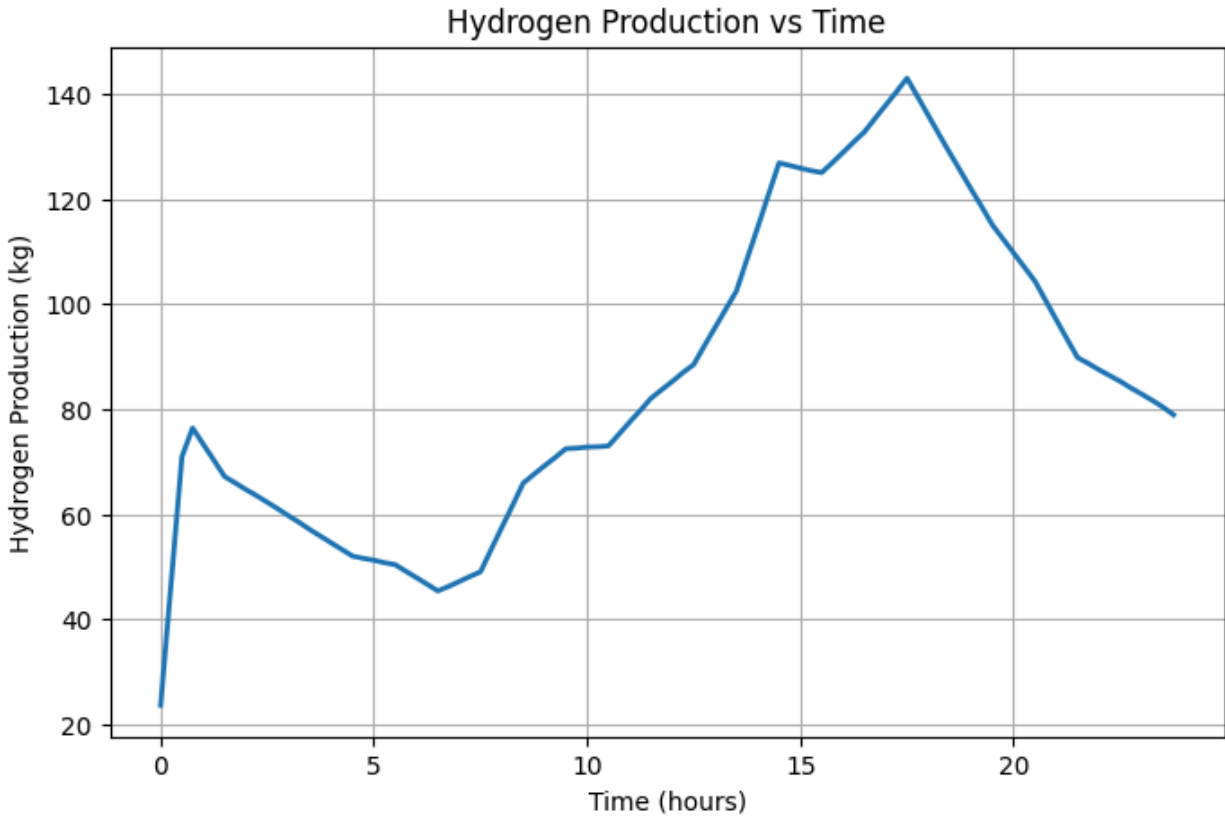


This graph shows the operation of the electrolyzer as wind power changes throughout the day. Initially, the wind power and electrolyzer power differ because the electrolyzer starts from the OFF state, and the ramp-rate constraint limits how quickly its power can increase. After the initial period, the electrolyzer power closely tracks the wind power because the available wind power remains within the electrolyzer's operating range.

Unlike solar power, wind power does not drop to zero during night hours and varies more continuously throughout the day. Therefore, the electrolyzer operates for a longer duration and produces hydrogen more continuously. In the solar model, the electrolyzer turned OFF during night because solar power became zero, while in the wind model, the electrolyzer can continue operating due to the continuous availability of wind power.



This graph shows the relationship between electrolyzer power and hydrogen production in the wind-based system. As the electrolyzer power increases, hydrogen production also increases almost linearly because more electrical energy is available for the electrolysis process. Compared to the solar case, the wind model operates at higher power levels for a longer duration, resulting in much higher hydrogen production. The graph is smoother and more continuous because wind power is available throughout most of the day.



This graph shows how hydrogen production changes over time as wind power becomes available. Initially, hydrogen production increases quickly as the electrolyzer starts operating. During periods of high wind power, hydrogen production increases because the electrolyzer receives more power. Unlike the solar case, hydrogen production does not become zero because wind power is available continuously throughout the day. Therefore, the wind-based system provides more stable, continuous hydrogen production than the solar-based system.

Conclusion

From both the solar and wind energy models, it can be observed that the PEM electrolyzer operation strongly depends on the availability of renewable power. In the solar-based system, hydrogen production occurs mainly during the daytime and ceases at night because solar power is unavailable. In contrast, the wind-based system provides more continuous, stable hydrogen production because wind power is available for a longer period throughout the day.

The solar system shows clear ON/OFF operation of the electrolyzer, while the wind system allows longer operating hours and higher hydrogen production. However, both systems still experience fluctuations in renewable power, leading to unstable electrolyzer operation and wasted excess energy. Therefore, a battery energy storage system is introduced to store excess renewable energy and supply power during periods of low generation. This helps reduce power fluctuations, improve electrolyzer stability, reduce renewable energy curtailment, and increase overall hydrogen production.